UNIT 3

Building a Quantum Computer - Theoretical Challenges & Requirements

Unit 3 at a glance

What is needed · fragility: decoherence, noise, control · conditions: isolation, error management, scalability, stability · why entanglement is hard to keep · error correction · hardware: superconducting, trapped ions, photonics · vision vs reality · role of quantum software

1. What is Required to Build a Quantum Computer?

A quantum computer is not a faster laptop. It is a machine that stores information in qubits, manipulates them with precisely timed physical operations, and reads out a result. David DiVincenzo (2000) listed the minimum requirements - now called the DiVincenzo criteria.

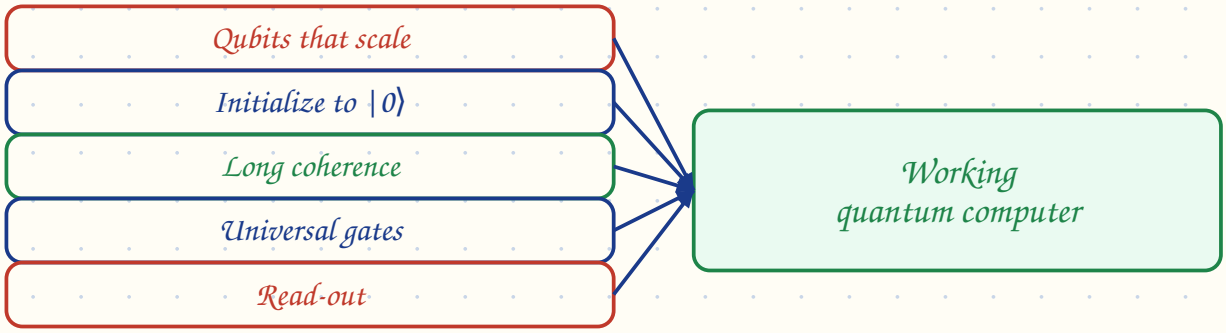


Fig 1: The DiVincenzo checklist

1. *Scalable physical system with well-defined qubits - many identical, well-characterized two-level systems.*
2. *Initialization - ability to reset all qubits to a known state such as $|000\ldots0\rangle$.*
3. *Long coherence times - far longer than the time of one gate operation.*
4. *Universal set of quantum gates - e.g. single-qubit rotations + one entangling two-qubit gate (CNOT/CZ).*
5. *Qubit-specific measurement - read out each qubit reliably.*
6. *(For networking) interconvert stationary and flying qubits; transmit them faithfully.*

★ *In one line*

Build good qubits, keep them quiet, control them precisely, read them out, and do it for millions of them.

» Ingredients in practice

Layer	What is needed
Qubits	Two-level quantum systems (circuits, ions, photons, spins)
Environment	Ultra-high vacuum or ~10-20 millikelvin refrigeration, magnetic shielding, vibration isolation
Control	Precisely shaped microwave / laser / voltage pulses with nanosecond timing
Read-out	Amplifiers, single-photon detectors, fast electronics
Classical computer	Real-time decoding, calibration, pulse sequencing
Software	Compilers, error-correction decoders, algorithms

2. Fragility of Quantum Systems

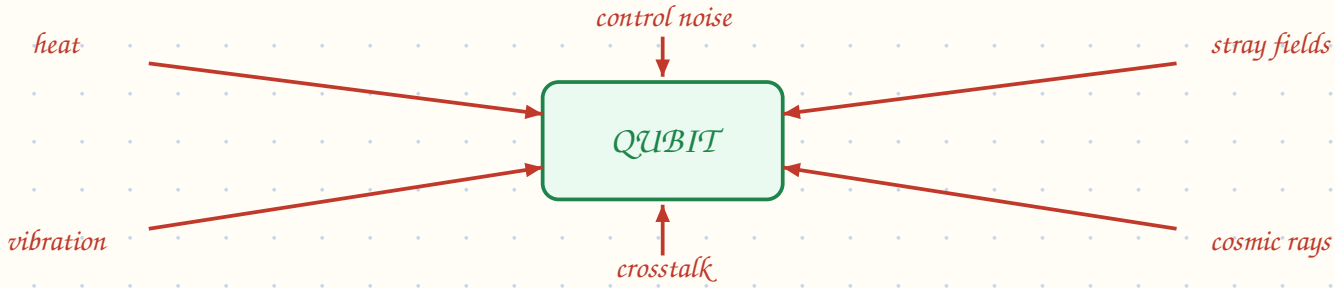


Fig 2: Everything in the environment attacks the qubit

» 2.1 Decoherence

The tiniest unwanted interaction acts like an unintended measurement and destroys superposition (see Unit 2). Sources: thermal vibrations, stray electromagnetic fields, defects in materials, cosmic rays, nearby qubits.

» 2.2 Noise

- Bit-flip noise: $|0\rangle \leftrightarrow |1\rangle$ by accident.
- Phase-flip noise: relative sign changes ($|+\rangle \leftrightarrow |-\rangle$) - unique to quantum, no classical counterpart.
- Amplitude damping: $|1\rangle$ loses energy and falls to $|0\rangle$ (T_1).
- Leakage: qubit escapes to a higher energy level outside the two levels.
- Crosstalk: control pulse for one qubit leaks to neighbours.

» 2.3 Control errors

Gates are analog operations: a rotation by an angle that is slightly off, or a pulse of slightly wrong length, produces a small error. Errors are continuous, accumulate over many gates and cannot be fixed by 'looking' (measurement would collapse the state).

Typical numbers (approximate, platform dependent)

Superconducting: gate ~20-100 ns, T_2 ~ 50-300 μs, 2-qubit error ~0.1-1 %. Trapped ion: gate ~ 10-100 μs, T_2 ~ seconds, 2-qubit error ~0.05-0.5 %.

A 1000-gate algorithm with 1 % error per gate has success probability ~ $(0.99)^{1000} \approx 0.004$ % - essentially zero. Hence error correction.

3. Conditions for a Functional Quantum System

Condition	Meaning	How achieved
Isolation	Protect qubits from environment but still allow control and read-out - a paradox: perfectly isolated qubits cannot be used	Vacuum, cryostats (10 mK), shielding, careful materials, suspended ions
Error management	Detect and correct errors faster than they accumulate	Quantum error correction codes, error mitigation, calibration
Scalability	Go from tens to millions of qubits without losing quality	Modular chips, interconnects, multiplexed control, 3-D wiring, cryo-CMOS
Stability	Keep device behaviour steady over hours/days	Automatic recalibration, temperature control, drift tracking

» Trade-offs - the 'quantum paradox'

- Want isolation (to stay quantum) vs want strong control (to compute).
- More qubits → more connections → more noise paths → lower quality per qubit.
- Faster gates give less decoherence per gate but need more powerful, noisier control.
- Cooling to millikelvin requires dilution refrigerators with limited wiring space - the 'wiring bottleneck'.

Exam tip / Remember

Remember the four conditions as I-E-S-S: Isolation, Error management, Scalability, Stability.

4. Theoretical Barriers: Why Maintaining Entanglement Is Difficult

- Entanglement is a collective property: the more qubits entangled, the more ways an environmental disturbance on any one of them ruins the whole state.
- Decoherence rate grows with system size - roughly linear in the number of entangled qubits (and even faster for 'cat-like' states, e.g. GHZ states).

- *Monogamy of entanglement: entanglement with the environment steals entanglement from your qubits.*
- *Entanglement is fragile to loss: for photons, a lost photon removes the entanglement entirely; fibre loss is exponential with distance (~0.2 dB/km).*
- *Gate imperfections: each entangling gate has the highest error rate (2-qubit errors ≫ 1-qubit errors).*
- *Long-range entanglement needs quantum repeaters because states can't be cloned or amplified.*

> *Entanglement lifetime picture*
> *If one qubit has coherence time T, N entangled qubits typically behave like they have T/N. A 100-qubit entangled state may last only 1/100 as long as one qubit!*

5. Error Correction as a Theoretical Necessity

Classical computers fix errors by redundancy: store 0 as 000, 1 as 111 and use majority vote. In quantum computing we cannot copy qubits (no-cloning) or look at them directly (collapse). Yet Peter Shor (1995) and Steane (1996) showed that quantum error correction (QEC) is possible!

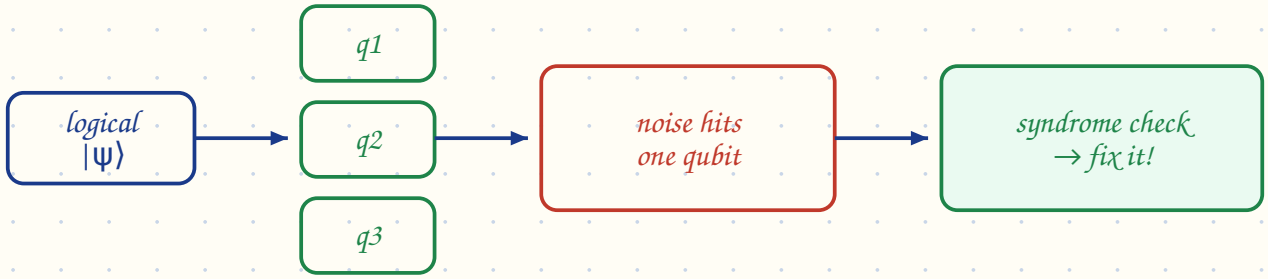


Fig 3: Idea of a quantum error-correcting code

» The key tricks

1. *Encode one logical qubit into many physical qubits using entanglement - not copying:* $\alpha|0\rangle+\beta|1\rangle \rightarrow \alpha|000\rangle+\beta|111\rangle$.
2. *Measure syndromes: parity checks on pairs of qubits (e.g. 'are q1 and q2 equal?') reveal WHERE an error happened without revealing* α, β.
3. *Correct by applying a fix gate to the flagged qubit.*
4. *Phase errors are handled by an additional code in the X-basis (Shor's 9-qubit code handles both).*

Threshold theorem

If the error rate per physical gate is below a critical value (the threshold, ~1 % for surface code), then by using more and more physical qubits the logical error can be made arbitrarily small. → Fault-tolerant quantum computing is possible in principle.

Code	Idea	Note
3-qubit repetition	Majority vote against bit flips	Teaching example
Shor 9-qubit	Handles any single-qubit error	First QEC code (1995)
Steane 7-qubit	CSS code from classical Hamming code	Good transversal gates
Surface code	2-D grid, local parity checks	Leading candidate; ~1000 physical per logical qubit
qLDPC / bosonic (cat, GKP)	More efficient encoding	Active research (IBM, Google, AWS)

- Overhead: a useful algorithm (e.g. RSA-2048 factoring) may need ~10^3-10^4 logical qubits → millions of physical qubits.
- Error mitigation (extrapolation, probabilistic cancellation) is used on NISQ devices without full correction - useful but not scalable.
- Milestone: Google (2023-24) showed logical error falling as code size grows (below threshold).

6. Quantum Hardware Platforms - Brief Conceptual Comparison

» 6.1 Superconducting circuits

Tiny electrical circuits made of aluminium/niobium on silicon, cooled below 20 mK where they become superconducting. A Josephson junction makes the circuit's energy levels uneven (anharmonic), so the lowest two levels act as a qubit (transmon). Used by IBM, Google, Rigetti, IQM, Alibaba, Fujitsu/RIKEN. Strengths: fast gates, lithographic fabrication like chips. Weaknesses: short coherence, fabrication variability, cryogenic wiring.

» 6.2 Trapped ions

Individual atoms (Yb^+, Ca^+, Ba^+) held in vacuum by oscillating electric fields; qubit = two internal levels; manipulated with lasers; entangled via shared vibrational motion. Used by IonQ, Quantinuum, Alpine Quantum Technologies. Strengths: identical qubits by nature, best gate fidelities (>99.9 %), all-to-all connectivity. Weaknesses: slower gates, scaling lasers & large traps (modular 'QCCD' architectures help).

» 6.3 Photonics

Qubits are photons (path/polarization/time-bin); gates via beam splitters, interferometers and detectors on silicon-nitride chips. Used by PsiQuantum, Xanadu, Quandela, USTC. Strengths: room-temperature operation (detectors may be cooled), natural for networking, low decoherence. Weaknesses: photon loss, probabilistic two-photon gates, need for many sources/detectors, large overhead.

Feature	Superconducting	Trapped ions	Photonics
Qubit	Circuit energy levels	Atomic levels	Photon modes
Temperature	~10 mK	Vacuum; ions laser-cooled; chamber ~room-T or 4 K	Room temp (detectors ~ 2-4 K)
Gate speed	Very fast (ns)	Slow (μs-ms)	Very fast (ps) but probabilistic
Coherence	~0.1 ms	seconds-minutes	limited by loss, not time
2-qubit fidelity	~99.5-99.9 %	~99.9 %+	depends on photon quality
Connectivity	Nearest neighbour (fixed)	All-to-all (in chain)	Flexible via waveguides
Scalability	Fab-friendly; wiring bottleneck	Trap/laser scaling	Manufacturable; loss management
Players	IBM, Google, Rigetti, IQM	IonQ, Quantinuum	PsiQuantum, Xanadu, Quandela

- Other: neutral atoms (QuEra, Pasqal, Atom Computing - 1000+ qubits arrays), silicon spin qubits (Intel, Diraq), NV centres, topological qubits (Microsoft's Majorana effort).
- Different platforms may win for different tasks - computing vs networking vs sensing; hybrid designs likely.

7. Vision vs Reality: What Works and What's Elusive

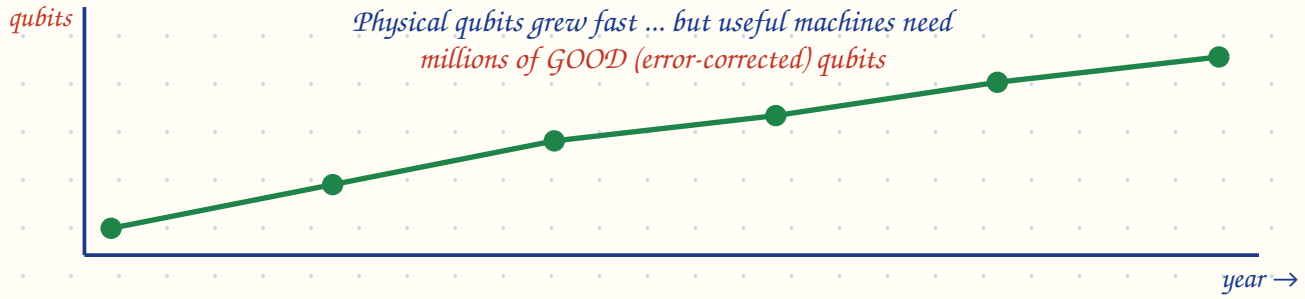


Fig 4: Qubit counts have exploded, but quality & error correction lag

Working today	Still elusive
Tens to ~1000+ noisy physical qubits (IBM Condor 1121; Atom Computing ~1200)	Large-scale fault-tolerant machines with 10^3-10^4 logical qubits

Working today	Still elusive
Quantum supremacy / advantage demos on contrived tasks (Google 2019 & 2024, USTC Jiuzhang)	Clear, practical advantage on useful commercial problems
Error-corrected logical qubits in small demonstrations (Google Willow, Quantinuum, QuEra/Harvard)	Reducing overhead of error correction (millions of qubits)
Cloud access to real devices (IBM Quantum, Amazon Braket, Azure Quantum)	Cheap, compact, reliable cryogenics & control wiring
QKD networks, quantum sensors, quantum random number generators	Quantum memory, repeaters and a global quantum internet
Algorithms known: Shor, Grover, quantum simulation, QAOA, VQE	More algorithms with clear exponential benefits

Honest summary

Hardware progress is real and accelerating, but a 'cryptographically relevant' or 'broadly useful' quantum computer is still years away (estimates range ~5 to 20 years). Hype must be balanced with realism - the 'quantum winter' risk if expectations are not met.

8. The Role of Quantum Software

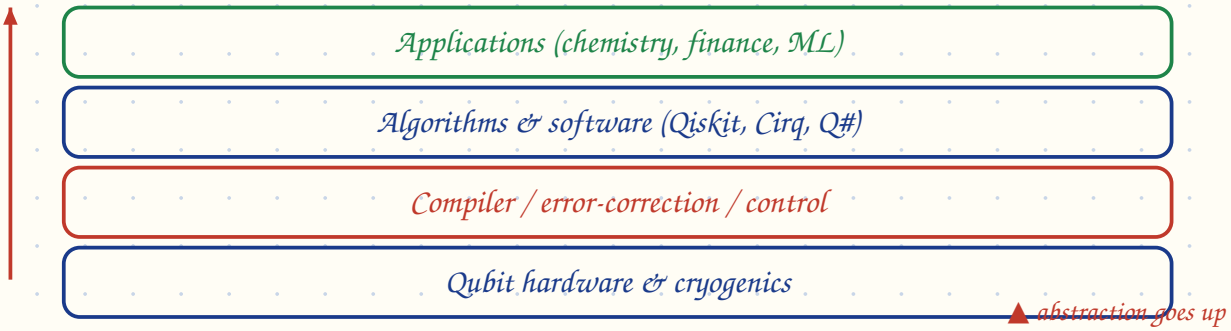


Fig 5: The quantum computing stack

Quantum software hides the physics and manages complexity so that algorithm designers need not control microwave pulses directly.

- *Programming frameworks*: IBM Qiskit, Google Cirq, Microsoft Q# / Azure Quantum, Xanadu PennyLane, Amazon Braket SDK, Quantum Development Kits.
- *Compilation / transpilation*: translate abstract circuits into the device's native gates, map logical qubits to physical ones respecting connectivity, minimize gates & depth.
- *Error mitigation & QEC software*: noise-aware compilation, zero-noise extrapolation, decoders that process syndrome data in real time.
- *Calibration & control software*: auto-tune pulses, track drift, schedule experiments.

- *Simulators & emulators: classical simulation of ~30-50 qubits for testing and education; noise models.*
- *Resource estimation: estimate the physical qubits and time needed for a target algorithm.*
- *Hybrid quantum-classical workflows: variational algorithms (VQE, QAOA) in which a classical optimizer tunes a quantum circuit; HPC integration.*

8A. Engineering-Free View of the Hardware Environment

» Why so cold? (superconducting & spin qubits)

Thermal energy kT must be much smaller than the qubit's energy gap hf. A transmon with $f \approx 5$ GHz has $hf/k \approx 0.24$ K, so we need $T \approx 10$-20 mK - colder than outer space (2.7 K). A dilution refrigerator reaches this using a mixture of helium-3 and helium-4, with stages at 50 K → 4 K → 800 mK → 100 mK → 10 mK. The 'chandelier' of gold-plated plates and coaxial cables is the iconic picture of a quantum computer.

» Why vacuum and lasers? (trapped ions, neutral atoms)

A single stray gas molecule colliding with an ion destroys its state, so ultra-high vacuum (~10^{-11} mbar) is used. Lasers cool the ions close to absolute zero (Doppler/sideband cooling), prepare them in a known state, drive gates, and make them fluoresce for read-out (bright = 1, dark = 0).

» Why precise light? (photonics)

Photon sources must emit one photon at a time, indistinguishable from each other; waveguides must have very low loss; detectors (superconducting nanowires) must be efficient. Because photons do not interact with each other, two-photon gates are built using measurement and ancillary photons ('measurement-based' or 'fusion-based' computing).

Type of noise	Cause	Remedy
Thermal	Not cold enough; hot wires	Better refrigeration, attenuation & filtering of lines
Magnetic/charge fluctuations	Material defects, two-level fluctuators	Cleaner fabrication, different qubit geometry (transmon), spin echo
Laser noise	Intensity/phase/frequency jitter	Stabilized lasers, composite pulses
Photon loss	Absorption, scattering, coupling losses	Better materials, heralded schemes, redundancy
Cosmic rays	High-energy particles break Cooper pairs → correlated errors	Shielding, underground labs, qubit design
Control electronics	Timing jitter, finite precision	High-quality DACs, calibration, cryo-CMOS

8B. Surface Code in Brief (Intuitive)

The surface code arranges data qubits on a checkerboard grid with measure qubits between them. Measure qubits repeatedly check 'parities' of their four neighbours. When an error occurs, some checks flip (a 'defect' appears); a classical decoder looks at the pattern of defects over time to guess the most probable error chain and correct it.

- *Needs only nearest-neighbour interactions on a 2-D chip - hardware friendly.*
- *Threshold ≈ 1 % (physical error rate must be below this).*
- *Larger code distance d → logical error falls roughly as $(p/p_th)^{(d+1)/2}$.*
- *Cost: d^2 data qubits per logical qubit (≈ $2d^2$ including measure qubits).*
- *Logical operations by 'lattice surgery' (merging/splitting code patches) or braiding defects.*

> *Fault-tolerance: three golden rules*
>
> *1. Error correction must not introduce more errors than it removes (below threshold). 2. A single fault must not spread to multiple qubits in one block. 3. Fresh, clean ancilla qubits are needed continuously - requires fast reset and measurement.*

8C. Key Challenges at a Glance

Challenge	Why it is hard	Active research
Qubit quality	Coherence/fidelity must be improved by another 10-100×	Materials science, new qubit types (fluxonium, bosonic)
Scaling	Wiring, power, cooling, crosstalk	Modular architectures, optical interconnects, cryo-CMOS
Error correction overhead	1000s of physical qubits per logical qubit	Better codes (qLDPC), better qubits
Connectivity	Nearest-neighbour limits circuits	Long-range couplers, shuttling ions, photonic links
Classical bottleneck	Decoding must keep up in real time (µs)	FPGA/ASIC decoders, AI decoders
Verification	How to check a result classical computers can't compute?	Cross-entropy benchmarking, interactive proofs
Workforce & cost	Interdisciplinary skills; expensive systems	Education, cloud access, national missions

★ *Remember*

Vision: million-qubit fault-tolerant computers solving chemistry and cryptography. Reality: few-hundred noisy qubits + first logical-qubit demos. Gap = error correction + engineering at scale.

9. Worked Examples

Example 1 - Gate budget

T_2 = 100 μs, gate time = 50 ns. Max operations before decoherence ≈ $100\times10^{-6} / 50\times10^{-9}$ = 2000 gates. Useful circuits needing 10^6 gates demand error correction.

Example 2 - Compound error

Per-gate fidelity 99.9 %, circuit with 500 gates: success ≈ 0.999^{500} ≈ 0.61 (61 %). With 99 % fidelity: 0.99^{500} ≈ 0.0066 (0.7 %). Shows why small improvements in fidelity matter greatly.

Example 3 - Overhead

If one logical qubit needs ~1000 physical qubits and an algorithm needs 4000 logical qubits, total ≈ 4 million physical qubits (+ ancilla/routing overhead).

10A. Benchmarking Progress: How Do We Measure a Quantum Computer?

Metric	Meaning
Number of physical qubits	Raw size - not enough by itself!
Gate fidelity	1 – error per gate; 2-qubit fidelity is the key number (≥ 99.9 % wanted)
Coherence times T_1, T_2	How long quantum information survives
Connectivity	Which qubits can directly interact
Quantum volume (IBM)	2^n where n = largest square random circuit (n qubits × depth n) run successfully
Algorithmic qubits (IonQ)	Number of usable qubits for algorithms
Randomized benchmarking	Estimates average gate error via long random gate sequences
Cross-entropy benchmarking	Compares sampling output with ideal simulation (Google)
Logical error rate	Error per logical operation after error correction - ultimate metric

★ *Lesson*

Ten high-quality qubits can outperform a hundred noisy ones. Always ask: how many qubits, how good, and how connected?

10B. Glossary for Unit 3

Term	Meaning
Transmon	Superconducting charge-insensitive qubit based on Josephson junction
Josephson junction	Two superconductors separated by thin insulator; gives nonlinear inductance
Fidelity	Closeness of actual state/gate to the ideal (1 = perfect)
Syndrome	Outcome of parity checks that signals which error occurred
Ancilla	Helper qubit used for measurement/error checks
Logical qubit	Encoded, error-protected qubit
Dilution refrigerator	Cryostat reaching ~10 mK using He-3/He-4 mixture
Crosstalk	Unwanted influence between neighbouring qubits/controls
Leakage	Qubit leaves the computational $\{\|0\rangle, \|1\rangle\}$ subspace
Fault tolerance	Computing reliably with unreliable components
Transpilation	Compiling a circuit to a device's native gates and connectivity

11. Short Answer Practice (Model Answers)

Q1. Why can't we just copy qubits for error correction?

No-cloning forbids copying an unknown state. Instead we spread the information of one qubit over an entangled state of many qubits ($\alpha\,|000\rangle + \beta\,|111\rangle$) and measure only parities (syndromes), which reveal errors but not α or β.

Q2. What is the difference between physical and logical qubits?

A physical qubit is the actual hardware two-level system (noisy). A logical qubit is an error-protected qubit encoded across many physical qubits, with much lower error rate. Useful algorithms are written in logical qubits.

Q3. Why do superconducting qubits need ~15 mK?

Thermal energy must be far below the qubit energy gap ($hf \approx 0.24$ K for 5 GHz) otherwise thermal photons randomly excite the qubit; also superconductivity of aluminium/niobium needs low temperature.

Q4. Compare quantum hardware in one line each

Superconducting - fast & fabricable; Ion - accurate & connected; Photonic - networkable & room-temperature; Neutral atoms - large arrays; Spin-silicon - small & CMOS-compatible; Topological - protected but unproven.

Q5. What is the threshold theorem and why is it important?

If physical error rate $p < p_{th}$, increasing code size reduces logical error exponentially. So arbitrarily long computations become possible with polynomial overhead - the theoretical justification for building large quantum computers.

Q6. What does 'quantum software' do that classical software does not?

It must respect reversibility and no-cloning, map circuits onto limited connectivity, schedule pulses, suppress noise, and run classical decoders in real time, while hiding hardware details from the programmer.

Q7. What is NISQ?

Noisy Intermediate-Scale Quantum: 50-1000+ imperfect qubits, no full error correction. Good for experiments, benchmarks and hybrid heuristics; not enough for Shor on large keys.

★ *Unit 3 summary in 5 lines*

1. Need scalable, initializable, coherent, controllable, measurable qubits. 2. Environment destroys quantum states → isolation + cold + vacuum. 3. Entanglement of many qubits is extra fragile. 4. QEC (surface code) with threshold makes large computers possible but costs ~1000× qubits. 5. Platforms and software are racing to close the gap between vision and reality.

12. Revision Questions

1. *Explain the DiVincenzo criteria.*
2. *Why are quantum systems fragile? Discuss decoherence, noise and control errors.*
3. *State the four conditions for a functional quantum system.*
4. *Why is it difficult to maintain entanglement?*

5. *Explain why quantum error correction is necessary and how it works without cloning.*
6. *Compare superconducting, trapped-ion and photonic platforms.*
7. *Distinguish between vision and reality in quantum computing today.*
8. *Describe the role of quantum software.*